

Report

Loss Function

The total loss is defined as:

$$\text{Total Loss} = \text{CrossEntropy Loss} + \lambda \times \text{Sparsity Loss}$$

Where:

- CrossEntropy Loss ensures classification performance
- Sparsity Loss is the average of all gate values

The sparsity loss encourages the network to deactivate unnecessary connections.

Why L1 Encourages Sparsity

The L1 penalty pushes parameters toward zero. Since gate values lie between 0 and 1, minimizing their sum encourages many gates to approach zero. This effectively removes connections, and makes the model sparse.

Results

Lambda	Test Accuracy	Sparsity (%)
0.001	0.4979	1.53
0.01	0.4945	2.05
0.1	0.4999	20.47

Observations

1. Increasing λ increases sparsity.
2. Higher sparsity reduces model accuracy.
3. There is a clear trade-off between model efficiency and performance.

Gate Distribution

The histogram of gate values shows:

- A large spike at 0: pruned connections

- A smaller spike at 1: important connections

